

Cell 0 — Markdown

Title + Notebook Overview

Title:

Demo: Multi-Agent Creative Story Building System

Description:

This notebook demonstrates how to design a **multi-agent storytelling system** where specialized agents collaborate to create high-quality fictional stories.

The system uses:

- Specialized creative agents
- Role-specific prompts
- Shared story memory
- Agent communication
- Iterative story improvement

Goal: Generate a coherent, creative story from a simple user idea.

Cell 1 — Markdown

Purpose

Explain the objective of the notebook:

This notebook demonstrates how to build a **creative writing multi-agent system** where agents collaborate to:

- Plan the story structure
- Design characters
- Build the world
- Write scenes
- Maintain consistency
- Improve quality

Example Input:

"A cyberpunk detective discovers memories can be stolen."

Example Output: A polished multi-chapter story with consistent plot, characters, and world-building.

Cell 2 — Code

Environment Setup (TODO)

TODO:

- Install required libraries
- Setup API keys
- Import dependencies

Planned libraries:

- LangChain
- LangGraph (optional)
- OpenAI / Anthropic model
- Memory utilities
- Structured output tools

TODO imports:

- LLM initialization
 - Agent framework
 - Prompt templates
 - Tool decorator
 - State management
 - JSON parser
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2. Defining Agent Roles

We will create a **Creative Story Team** with specialized responsibilities.

Agents:

1. Story Planner Agent

Responsibilities:

- Understand story idea
- Decide genre & tone
- Create plot outline
- Define beginning, middle, climax, ending

Output: High-level story blueprint

2. Character Designer Agent

Responsibilities:

- Create main characters
- Define motivations
- Character flaws
- Backstories
- Relationships

Output: Character profiles

3. World Builder Agent

Responsibilities:

- Design story setting
- Rules of the world
- Locations
- Culture
- Technology / magic systems

Output: World-building document

4. Scene Writer Agent

Responsibilities:

- Write scenes
- Convert outline into narrative
- Add dialogue
- Maintain pacing

Output: Story chapters/scenes

5. Consistency Checker Agent

Responsibilities:

- Detect contradictions
- Check plot holes
- Ensure character consistency
- Maintain timeline continuity

Output: Story fixes & revisions

6. Style Editor Agent

Responsibilities:

- Improve writing quality
- Enhance dialogue
- Improve emotional impact
- Fix awkward wording

Output: Polished final story

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Imports + Tool Definitions (TODO)

TODO: Create custom tools for storytelling.

Planned Tools:

Tool 1: Genre Inspiration Tool

Purpose: Provide storytelling inspiration based on genre.

Example: Fantasy Sci-Fi Mystery Horror

Returns: Themes, tropes, ideas

Tool 2: Character Name Generator Tool

Purpose: Generate fitting names based on:

- culture
- genre
- personality

Returns: Character names

Tool 3: Plot Twist Generator Tool

Purpose: Suggest unexpected twists.

Returns: Creative plot twist ideas

Tool 4: Story Memory Tool

Purpose: Store:

- characters
- locations
- story facts
- important events

Prevents contradictions.

Tool 5: Lore Lookup Tool

Purpose: Allow agents to search previously created story information.

Used for consistency.

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Initialize LLM (TODO)

TODO: Initialize model.

Define:

- temperature for creativity
- max tokens
- reasoning level
- model configuration

Experiment ideas:

- High creativity model
 - Balanced model
 - Editor model with lower temperature
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3. Role-Specific Prompts

Explain why specialized prompts matter.

Each agent will have:

- Clear responsibility

- Strict boundaries
- Specific output format

Goal: Avoid overlapping responsibilities.

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Story Planner Prompt (TODO)

Prompt responsibilities:

You are a Story Planner Agent.

Your ONLY job:

- Analyze user story idea
- Create narrative structure

Rules:

- Do not write full scenes
- Focus on plot only
- Include:
 - Act 1
 - Act 2
 - Act 3
 - climax
 - ending
 - conflict

Output format: Structured outline

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Character Designer Prompt (TODO)

Prompt responsibilities:

You are a Character Designer Agent.

Your ONLY job: Design memorable characters.

Include:

- name
- personality

- strengths
- flaws
- goals
- fears
- relationships
- character arc

Rules:

- Avoid clichés
- Match story genre

Output: Character sheet

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World Builder Prompt (TODO)

Prompt responsibilities:

You are a World Builder Agent.

Your ONLY job: Create immersive settings.

Include:

- locations
- society
- rules
- environment
- political systems
- technology/magic

Rules:

- Maintain internal logic

Output: World specification

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Scene Writer Prompt (TODO)

Prompt responsibilities:

You are a Scene Writer Agent.

Your ONLY job: Write scenes using:

Inputs:

- story outline
- characters
- world data

Requirements:

- natural dialogue
- emotional pacing
- vivid descriptions
- coherent transitions

Rules:

- Do not change plot

Output: Story chapter

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Consistency Checker + Style Editor Prompts (TODO)

Consistency Checker

Responsibilities:

- Find plot holes
- Timeline mistakes
- Character inconsistencies
- Broken world rules

Output: Corrections list

Style Editor

Responsibilities:

- Improve prose
- Improve readability
- Strengthen emotions
- Enhance dialogue

Rules:

- Keep author's original meaning

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4. Creating Role-Based Agents

Explain agent construction.

TODO: Create agents:

- `planner_agent`
- `character_agent`
- `world_builder_agent`
- `scene_writer_agent`
- `consistency_checker_agent`
- `style_editor_agent`

Each agent gets:

- specific tools
 - dedicated prompt
 - separate responsibilities
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Create Agent Executors (TODO)

TODO: Build executors for each agent.

Map tools:

Story Planner:

- Genre Inspiration Tool

Character Designer:

- Name Generator Tool

World Builder:

- Lore Tool

Scene Writer:

- Story Memory Tool

Consistency Checker:

- Lore Lookup Tool

Style Editor:

- Editing utilities
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Cell 14 — Code

Multi-Agent Story Workflow (Main Pipeline) (TODO)

Function: `run_story_builder(user_prompt)`

Workflow:

STEP 1: Planner Agent → Creates story outline

STEP 2: Character Agent → Creates characters

STEP 3: World Builder Agent → Creates world

STEP 4: Scene Writer Agent → Writes story chapter-by-chapter

STEP 5: Consistency Checker → Detects issues

STEP 6: Scene Writer revises sections

STEP 7: Style Editor → Produces polished story

STEP 8: Final Output → Return complete story

Add logging:

- Agent actions
 - Inputs/outputs
 - revisions
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5. Reflection Questions

1. Why do creative systems benefit from specialized agents?
2. Which agent had the biggest impact on story quality?

3. What problems arise without consistency checking?

4. How can memory improve long stories?

5. What new agents could improve storytelling?

Examples:

- Dialogue Agent
 - Villain Agent
 - Emotion Agent
 - Humor Agent
 - Plot Twist Agent
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Testing Section (TODO)

Example prompts:

1. "Write a dark fantasy story about a cursed king."
2. "A detective discovers AI can rewrite memories."
3. "A survival story on Mars with psychological horror."

TODO: Run pipeline and inspect outputs.